Trees Data Structure — Non-Linear Data Structure

Graphs

Data Structures and Algorithms
Binary Tree

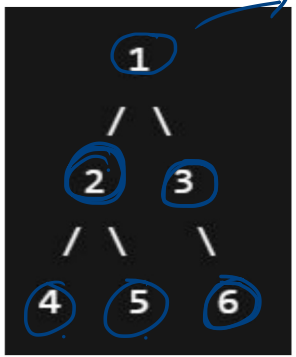
Mentor: **Adarsh Gupta**
(Senior Software Engineer)

Binary Tree:

A Binary Tree is a type of tree data structure in which each node can have at most two children — referred to as the left child and right child.

0, 1, 2 ✓

not more than 2

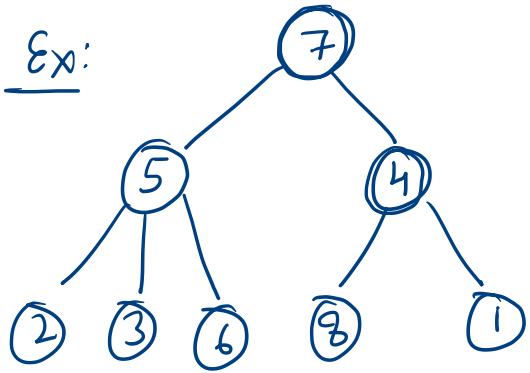


root node

leaf nodes = 4, 5, 6

B. T.

Ex:



No

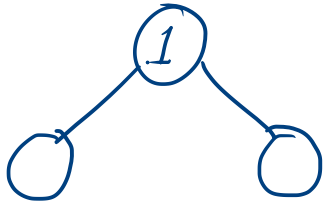
How to create Tree Node in Java

```
class Node {
    int data;
    Node left, right;

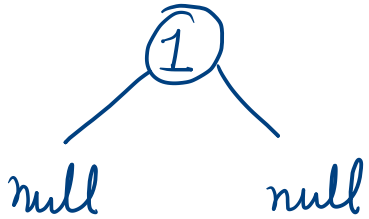
    Node(int value) {
        data = value;
        left = right = null;
    }
}
```

1

// constructor



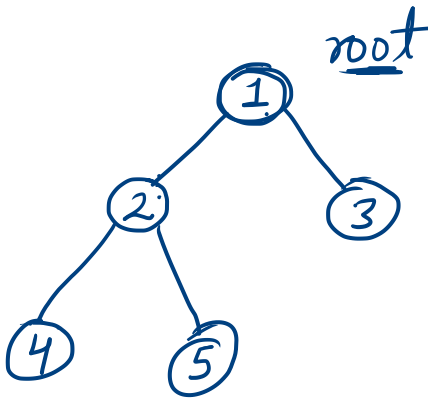
data = 1



Binary Tree Implementation in Single Main Class

```
public static void main(String[] args) {
    Node root = new Node(1);
    root.left = new Node(2);
    root.right = new Node(3);
    root.left.left = new Node(4);
    root.left.right = new Node(5);

    System.out.println("Binary Tree created successfully.");
}
```



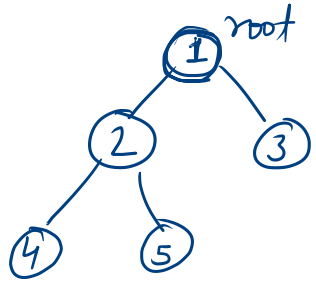
Better Structure way to implement a Binary Tree

```
class Node {

}
```

```
class BinaryTree {
    Node root;
    BinaryTree() {
        root = null;
    }

    void createTree() {
        root = new Node(1);
        root.left = new Node(2);
        root.right = new Node(3);
        root.left.left = new Node(4);
        root.left.right = new Node(5);
    }
}
```




```
public class Main {
    public static void main(String[] args) {
        BinaryTree tree = new BinaryTree();
        tree.createTree();
        sout("Binary Tree created");
    }
}
```

Advantages of using multiple classes:

- **Encapsulation**: Node and Tree logic are separated cleanly.
- **Scalability**: Easy to add more tree operations later (e.g., search, insert, delete).
- **Reusability**: The Node class can be reused for Binary Search Trees or AVL Trees later.
- **Modularity**: Each class has its own responsibility.

Most Imp.

Tree Traversal Techniques → Concept → solve Problem → Code

☆ Revise the Recursion topic → Leap of faith